



Optical fiber tip integrated photoelectrochemical sensors

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Abstract: In this work, we design and fabricate a compact photoelectrochemical (PEC) sensor by integrating a graphene-MoS₂ heterostructure on an optical fiber tip. The graphene serves as a transparent carrier transport layer, and the MoS₂ presents a photoelectrical transducer that generates photocarriers and interacts with ascorbic acid (AA) in solution. This device is used to demonstrate a self-powered detection of AA with a concentration range between 1 mM and 50 mM, and a time response of ~ 6 ms. The device downsizes traditional PEC systems to the micrometer scale, benefiting the real-time monitoring of biochemical changes in small areas and opening the pathway for miniaturized PEC sensing applications.

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1. Introduction

In recent years, photoelectrochemical (PEC) sensing shows great potential in the field of biochemical molecular detection, which is self-powered, low-cost, and fast response. Different from traditional electrochemical and optical methods, the excitation source (light) and detection signal (current) of PEC sensing are separated, benefiting from the low background signal and high sensitivity to trace amounts of molecules. By combining with biological recognition elements (enzymes [1–4], nucleic acids [5–8], antigens/antibodies [9–13], etc.), PEC sensors also show great selectivity. Nevertheless, the large-sized, separated structure of a conventional PEC sensing system impedes its further miniaturized applications.

Optical fibers, as the most common light waveguides, are great candidates for flexible optoelectronic functional integration. Different materials and microstructures [14] have been integrated onto optical fiber endfaces to overcome the limited material properties of optical fibers (most often SiO₂). Recently, two-dimensional (2D) materials have shown great potential in optical sensing field due to their excellent opto-electromechanical properties, which are ideal candidates for fiber platform integration. To date, a series of fiber-integrated optoelectronic devices have been achieved, such as photodetectors [15–18], modulators [19,20], lasers [21,22] and thermal detectors [23,24] and biosensors [25–27].

Ascorbic acid (AA), known as vitamin C, is a critical substance for human health. The insufficient intake of AA may cause a variety of diseases, such heart failure, bleeding gums, mental illness and so on [28]. So far, many strategies have been developed for AA detection, including electrochemistry [29], colorimetry [28,30] chemiluminescence [31], SPR [25–27] and others. Most of the methods need complex manufacturing process and expensive test equipment.

In this work, we have constructed a highly integrated all-fiber PEC sensing platform, which combines the working and counter electrodes on fiber tips. For application demonstration, we integrate a graphene and molybdenum disulfide (MoS₂) heterostructure on a multimode fiber (MMF) tip to fabricate a compact PEC sensor. In the heterostructure, graphene is the transparent carrier transport layer, and MoS₂ presents a photoactive layer that generates photocarriers and interacts with target molecules. The device exhibits a sensitive response to AA in phosphate

buffer solution (PBS), with a detection limit of ~ 1 mM under 400 nm illumination and a fast response time of ~ 6 ms. Thanks to the self-powered property of the PEC sensor, the devices can work without an external bias voltage supply. The device downsizes traditional PEC systems to the micrometer scale, benefiting the real-time monitoring of biochemical changes in a small area. Furthermore, the miniaturized, flexible, low-cost, fiber-integrated device may offer a new strategy for PEC sensing applications for human body monitoring, wearable electronics, and other special detection scenes.

2. Device design

In general, PEC sensing refers to the influence of analyte concentration on the current signal, which involves charge transfer between the photoactive material and analyte. As shown in Fig. 1(a), a pair of gold electrodes on the endface and sidewall of an MMF serve as working and counter electrodes, respectively, where the redundant gold film was removed by tapered tungsten needle (endface) and lapping films (sidewall). The working electrode at the endface connects to a transferred graphene-MoS₂ heterostructure, which is isolated to the counter electrode. The transparent graphene [32] layer serves as an extent of the working electrode, ensuring high carrier transport efficiency and high light transmission. MoS₂ [33,34] is the key photoactive layer that generates photoexcited electron-hole pairs and interacts with AA in solution, which is displayed in Fig. 1(b). Under illumination, MoS₂ is photoexcited to generate electron-hole pairs. The AA molecules in PBS act as electron donors and combine with the holes in MoS₂ to cause electron accumulation at the working electrode. The potential difference is built between the working and counter electrodes, thereby generating photocurrent in the external circuit, which is related to AA concentration. Due to the graphene-MoS₂ heterostructure, the electron holes can be separated efficiently, which is beneficial for improving the sensitivity and response speed of our device. In this way, we can directly calibrate the concentration of AA according to the change in photocurrent.

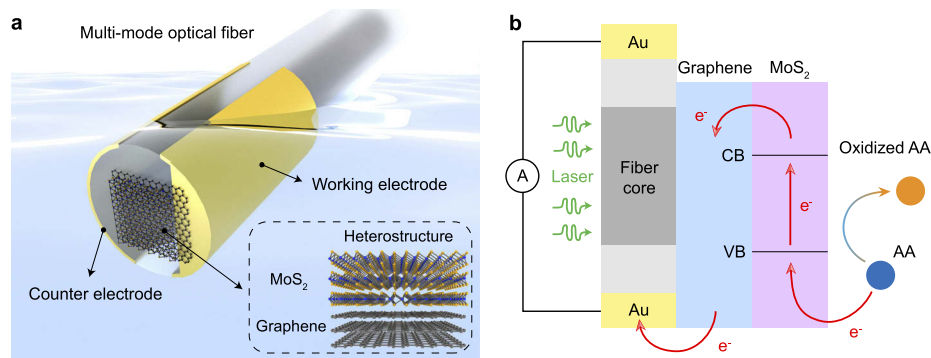


Fig. 1. Device design and principle. **a**, Schematic of the device. The PEC sensing part of the device is integrated with graphene and MoS₂ film on the fiber endface. **b**, Photocurrent generation mechanism of the device for the detection of AA in PBS.

3. Fabrication and test

3.1. Device fabrication and characterization

On the basis of our previous work [15–17], we designed and fabricated the all-fiber PEC sensing device by the sequential fabrication process shown in Fig. 2. First, a multimode optical fiber (core: 62.5 μ m, cladding: 125 μ m, Yangtze optical fiber company, YOFC) was cleaned by acetone, ethanol, and deionized water sequentially, and then an ~ 30 nm thick gold film was deposited

uniformly covering the entire fiber. The purpose of choosing MMF is to enhance the transmitted light power and enlarge the light-matter interaction area, which is beneficial to improve the photocurrent and sensitivity of the device. Then, we employed a tapered tungsten needle to scratch the gold layer at the center of the fiber endface to form a channel, which was carried out under an optical microscope with a high precision moving stage. Note that the width of the channel fabricated on the endface was approximately 70 to 80 μm to completely expose the core of the MMF. Then, two pieces of lapping films were used to remove a portion of the gold layer at the lateral wall of the fiber and left a gap between two electrodes. After that, copper-based chemical vapor deposition (CVD)-grown graphene (Six Carbon Technology) and CVD-grown MoS_2 on mica (Six Carbon Technology) were transferred to the fiber endface sequentially by a dip-coating method [35]. Considering the transmittance and electrical conductivity of graphene, the graphene we used is about 8~10 layers. The MoS_2 film we used is about 5 layers, due to the high mechanical strength, large absorption and low defects density. The sample was baked at 180 $^{\circ}\text{C}$ for 5 min to improve the contact between the material and substrate after transfer. Finally, we used the tapered tungsten needle again to scratch off the unnecessary area of materials on the fiber endface, ensuring that the two electrodes were disconnected completely.

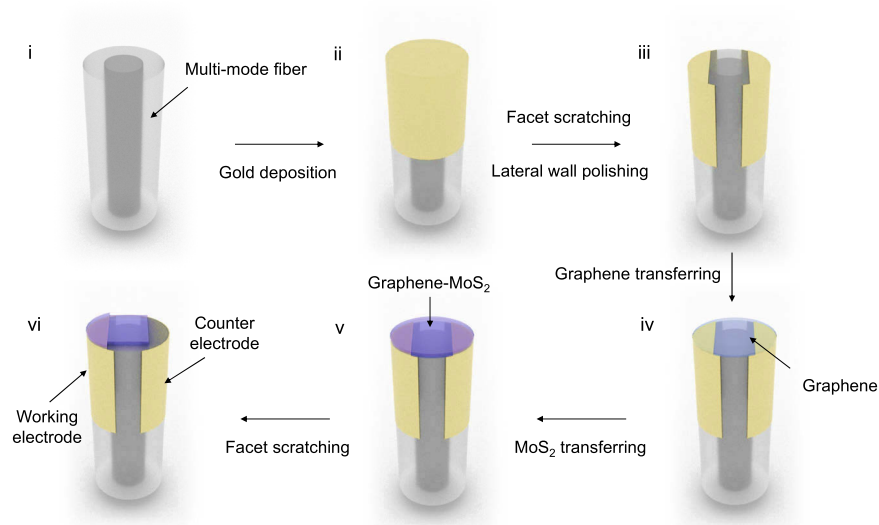


Fig. 2. Sequential fabrication process of the device.

The colored scanning electron microscopy (SEM) image of a prepared device is shown in Fig. 3(a). The white dashed circle represents the core of the MMF, which is the functional area for PEC sensing under illumination. The transmission curve of the graphene- MoS_2 heterostructure was measured and is shown in Fig. 3(b). The heterostructure shows high absorption in the visible spectrum, which is related to the band gap of MoS_2 . Figure 3(c) shows the Raman spectra of the transferred graphene- MoS_2 heterostructure. There are two typical vibration modes of MoS_2 in the low-frequency region, and the E_{12g}^1 (Γ) and A_{1g} (Γ) modes are identified at 373.8 cm^{-1} and 400.7 cm^{-1} . In the high-frequency region, three characteristic peaks of graphene at 1350.8 cm^{-1} (D-band), 1572.1 cm^{-1} (G-band) and 2691.9 cm^{-1} (2D-band) are observed. SEM-EDS was also presented to demonstrate the detail material composition of our device, which is shown in Fig. S1 in Supplement 1.

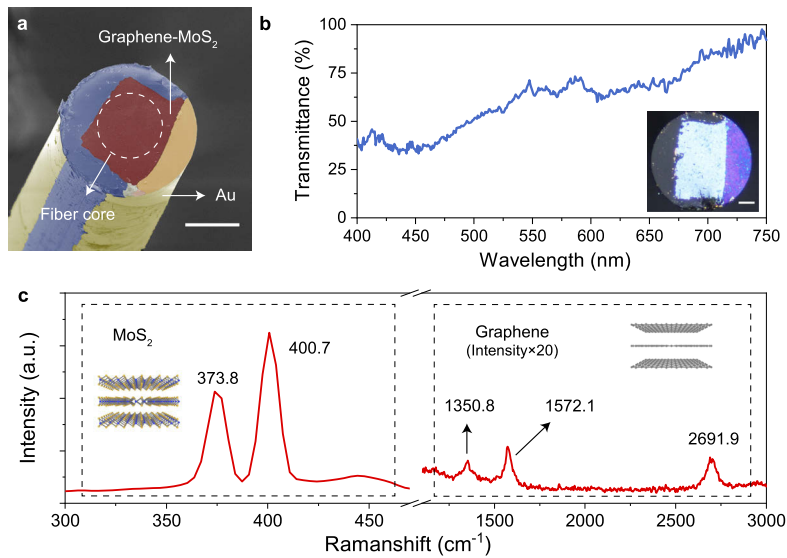


Fig. 3. Device characterizations. **a**, Colored SEM image of the device, where the white dashed circle represents the fiber core. The scale bar is 50 μm. **b**, Transmission curve of the multilayer graphene-MoS₂ heterostructure. **Inset**: Microscopic image of a prepared device. The scale bar is 20 μm. **c**, In situ Raman spectra of the multilayer graphene-MoS₂ heterostructure under 532 nm excitation.

3.2. Detection of the AA concentration

The fiber tip of the device was immersed in PBS [1,2,36,37], which is the superior solvent for AA molecules [26], and the input light ($\lambda = 400$ nm) was coupled into the MMF and transmitted to the PEC sensing area directly. In addition, a reference optical power meter was used to calibrate the input light power, and a digital source meter (Keithley 2450, Tektronix) was used for the measurement of photocurrents, which is connected to the working and counter electrodes on MMF through a miniaturized electrode holder.

Figure 4 shows the photocurrent (defined as $I_p = I_{\text{light}} - I_{\text{dark}}$) versus different AA concentrations and light powers. We explored the impact of incident light power on the photocurrent in PBS with 50 mM AA, and the experimental results are shown in Fig. 4(a) and Fig. 4(b). Significant photocurrents were generated under illumination, and the photocurrents showed a trend of approaching saturation with increasing illumination intensity.

Then, we investigated the relationship between the concentration of AA and the value of photocurrent under a fixed incident light power of 3 mW. The concentration range of AA was changed by adding prepared AA solution (dissolved in PBS, 0.1 M) to pure PBS step by step. After each addition, the PBS was rested for 15 minutes to ensure the uniformity of the solution. As shown in Fig. 4(c), obvious photocurrents were observed even at low AA concentrations of 5 mM, and huge differences in photocurrents were presented with different AA concentrations. The relationship between photocurrents and AA concentrations is shown in Fig. 4(d), showing a trend of approaching saturation with increasing illumination intensity at a high concentration range of 10 to 50 mM. Under a low concentration range of 1 to 5 mM, the device showed a linear photocurrent-concentration relationship with a sensitivity of 0.17 nA/mM.

3.3. Photoswitching performance of the device

To confirm the photoswitching characteristics and stability of our device, the currents under a periodically switched light source were investigated, as shown in Fig. 5. The photocurrent

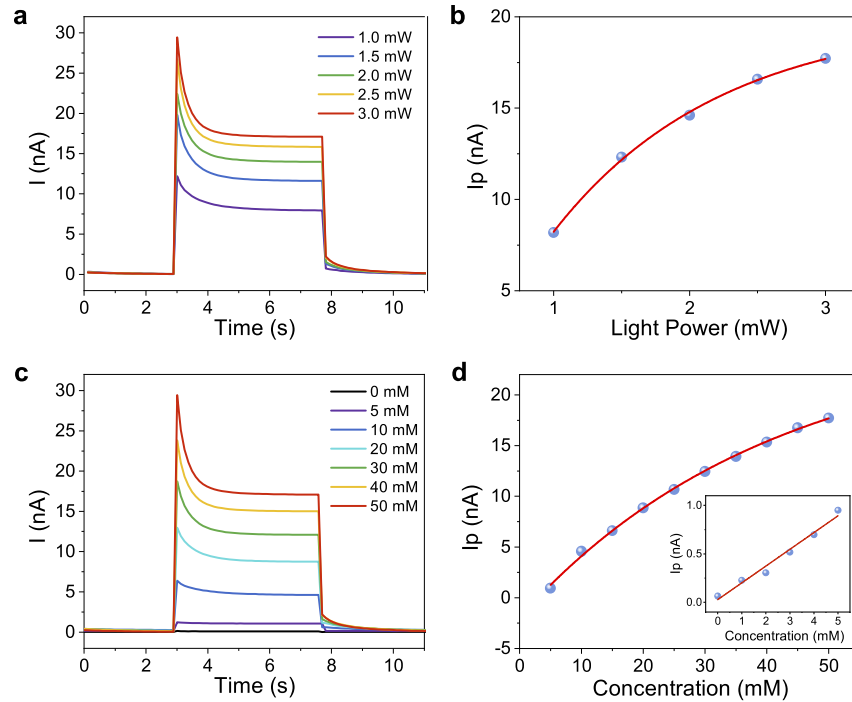


Fig. 4. Photocurrent characterization of the device. **a**, Dynamic current curves under different light powers ($\lambda = 400$ nm) and with 50 mM AA in PBS. **b**, Photocurrents are related to light powers. **c**, Dynamic current curves in the presence of different amounts of AA, which was operated at a light power of 3 mW. **d**, Photocurrents are related to AA concentrations. **Inset**: the photocurrent generated with 1-5 mM AA concentrations.

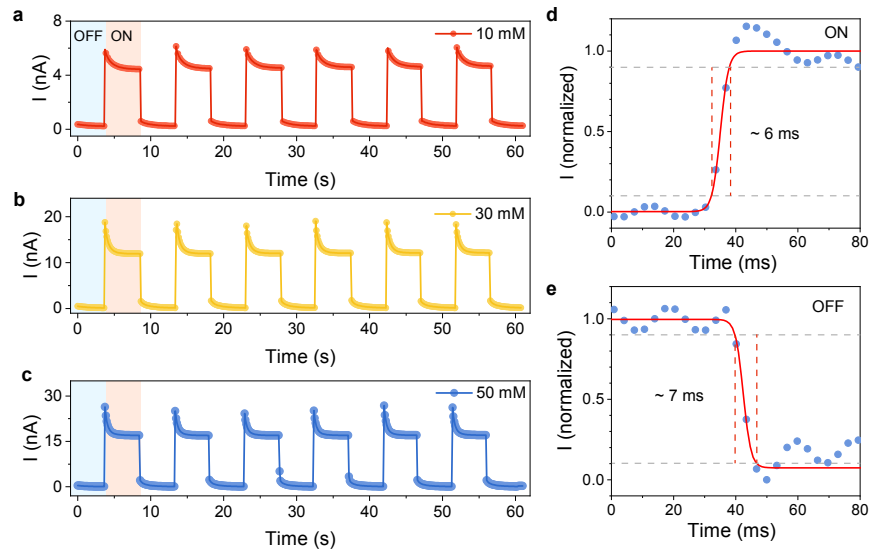


Fig. 5. Photocurrent dynamics. **a-c**, Photocurrent dynamics of the device under different AA concentrations, with a fixed light power of 3 mW. **d** and **e**, Enlarged views of the photocurrent dynamics during one cycle of light modulation with 30 mM AA in PBS.

dynamics were almost identical for different concentrations of AA. The currents increased immediately when light was turned on and decreased to nearly zero when the light was turned off. In addition, we applied the ON-OFF switches of the light 6 times, as depicted in Fig. 5(a)–5(c), showing the high stability for the switching behavior of the device. Due to the high carrier mobility of graphene and the quick consumption of photoinduced carriers when the reaction between AA and MoS₂ was suspended, the photocurrent changed immediately when the light source was switched, with rise/fall times of 6/7 ms under 30 mM AA, as shown in Fig. 5(d) and 5(e). It is notable that strict cleaning process is required after each usage, due to the sensitive response of the 2D materials to impurities.

4. Conclusion

In summary, we demonstrate a compact photoelectrochemical sensor by integrating a graphene-MoS₂ heterostructure on an MMF tip. Compared to traditional PEC sensors, our fiber-integrated PEC sensor is highly integrated, flexible and cost-effective, benefiting the real-time monitoring of biochemical changes in small areas. Thanks to the internal built-in field of the graphene-MoS₂ heterostructure for carrier separation, the device shows self-powered detection of AA with a concentration range between 1 mM and 50 mM, with a time response of ~ 6 ms. The performance may be further improved by several attempts, such as using larger-core fibers, using better-quality photoelectrical transducer, optimizing the electrode structure, performing signal processing and so on. Due to the lack of recognition elements, our device is temporarily not selective for molecules detection, which can be achieved by designing PEC reactions and integrating various recognition elements on fiber tips in future research. After above improvements, our all-fiber PEC platform may play an important part in human body monitoring, wearable electronics, and other special detection scenes.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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